SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY ONE)
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. A system has 20 MHz of spectrum with 25 kHz simplex channels. Derive the number of full-duplex FDMA channels, showing the spectrum-to-simplex-to-duplex conversion step by step. Explain why full-duplex operation requires pairing two simplex channels, and where the frequency separation (uplink/downlink) typically comes from in a real system. Conclude by discussing how this channel count directly limits the number of simultaneous users the system can serve.
2. A TDMA frame lasts 4.615 ms with 8 slots, 2 of which are reserved for signaling. Derive the slot duration and the number of usable traffic slots, then use this to derive the total traffic capacity across 60 carriers. Explain why signaling slots are necessary even though they reduce user capacity. Discuss what would happen to voice quality and system control if signaling slots were removed to maximize traffic capacity.
3. Derive the number of carriers and the total number of voice channels for a GSM system given 12.4 MHz of spectrum, 200 kHz carrier spacing, and 8 time slots per carrier. Show the full calculation chain from spectrum to carriers to channels. Discuss what portion of this theoretical maximum a real deployment would actually achieve once signaling and guard-band overhead are accounted for, and why that gap matters for capacity planning.
4. A system switches carrier spacing from 200 kHz to 100 kHz while keeping the same 15 MHz allocation. Derive the resulting change in the number of carriers and the corresponding voice-channel capacity (8 slots/carrier), showing before-and-after values clearly. Explain the interference and receiver-selectivity trade-offs introduced by halving the carrier spacing, and discuss why this isn't simply a "free" capacity gain in practice.
5. Derive the total number of usable FDMA channels for a 30 MHz spectrum allocation with 25 kHz per channel, after accounting for a 10% bandwidth loss to guard bands. Show the calculation for both the theoretical channel count (without guard-band loss) and the net usable count. Discuss why guard bands are a necessary "tax" on capacity, and how operators try to minimize this loss through tighter filtering.
6. A TDMA system must support 480 users using 8 slots per carrier. Derive the number of carriers required, and then the total spectrum needed at 200 kHz per carrier. Present your derivation with intermediate steps clearly shown. Discuss how a network planner would use this exact calculation when sizing a new cell site for a projected subscriber base, and what safety margin might be added in practice.
7. Derive the new slot duration if TDMA frame length increases from 4.615 ms to 6.0 ms while the number of slots per frame stays at 8, showing the before-and-after slot durations. Discuss the implications of this longer slot duration for latency-sensitive voice traffic, including potential effects on perceived call quality, echo, and synchronization requirements. Explain why standards bodies chose a specific frame length rather than a longer one.

8. A hybrid FDMA/TDMA system allocates 25 carriers × 8 slots each. If 15% of the total slots are reserved for signaling, derive the net number of channels available for actual user traffic, showing the full calculation. Discuss why this signaling overhead exists even in a "hybrid" system design, and how it compares proportionally to signaling overhead in a pure FDMA or pure TDMA system.
9. Derive the total control-channel bandwidth for a system with 150 control channels at 10 kHz each, and then calculate what percentage this represents of a 25 MHz total spectrum allocation. Show both calculations with clear steps. Discuss why this percentage matters to an operator — i.e., what this "signaling tax" costs them in terms of foregone revenue-generating capacity — and how it might be reduced through more efficient control-channel design.
10. Compare FDMA (30 kHz/channel) and TDMA (200 kHz/carrier, 8 slots) for the same 12 MHz spectrum allocation. Derive the total channel capacity for each scheme, showing full working for both. Present the resulting capacity numbers side by side, and justify the spectral-efficiency gap between the two approaches — explaining specifically why TDMA achieves more channels per MHz despite requiring wider individual carrier bandwidth.
11. A CDMA system has a chip rate of 1.2288 Mcps and supports 9.6 kbps voice. Derive the processing gain in both linear and dB form, showing the full calculation and unit conversion. Explain what this processing-gain value physically means for the receiver's ability to recover a signal buried in noise/interference. Discuss why voice CDMA systems are typically designed around this specific chip rate.
12. Derive the maximum user data rate for a spreading factor of 128 at a chip rate of 3.84 Mcps, then repeat the derivation for a spreading factor of 64, showing both calculations clearly. Compare the two resulting data rates and explain the underlying trade-off: lower spreading factor gives higher data rate but reduced processing gain and interference robustness. Discuss which spreading factor a network might choose for voice versus data services.
13. A Direct-Sequence Spread Spectrum (DS-SS) system uses a 32-chip spreading code. Derive the processing gain from this code length, and then derive the chip rate required to support a 50 kbps data rate at this spreading factor. Show both calculations with clear working. Discuss how this relationship between spreading code length, chip rate, and data rate constrains system design choices in practice.
14. Derive the processing gain in dB for a system with a 5 Mcps chip rate and a 12.2 kbps data rate, showing the full linear-to-dB conversion. Explain, using this derived value, why a higher processing gain allows a signal to be transmitted and successfully recovered even when its power is below the ambient noise floor. Discuss the practical interference-resistance and security benefits this property historically provided.
15. A CDMA system has a processing gain of 256 and requires a linear SIR of 6. Using SIR = PG/(N−1), derive the maximum number of simultaneous users the system can support, showing the algebraic rearrangement step by step. Discuss why this relationship makes CDMA capacity "soft"

(interference-limited) rather than hard-limited like FDMA/TDMA channel counts, and what this means operationally.

16. Derive the new maximum-user count when processing gain doubles from 128 to 256 while the required linear SIR stays fixed at 8, showing both before-and-after calculations clearly. Discuss why a network operator might deliberately increase processing gain — through a higher chip rate or reduced data rate — specifically to boost user capacity, and what real-world costs (spectrum, hardware) this trade-off imposes.
17. A CDMA system requires Eb/N0 = 5 dB. Derive the required SIR in dB for a processing gain of 128, using SIR ≈ Eb/N0 − 10log10(PG), showing every step of the conversion and subtraction. Explain why this relationship between Eb/N0 and required SIR is a central design target in CDMA engineering, and discuss what a lower required Eb/N0 (via better coding) would mean for system capacity.
18. Derive the effective interference reduction when a CDMA system's voice-activity factor drops from 1.0 to 0.45, showing the calculation clearly. Then discuss its impact on the number of supportable users in a 3-sector CDMA cell, explaining how voice-activity averaging and sectorization compound together to increase effective capacity beyond a naive worst-case estimate.
19. Derive the processing gain for a system with a 3.6864 Mcps chip rate at three different data rates — 9.6, 19.2, and 38.4 kbps — showing all three calculations clearly side by side. Comment on the trend you observe (processing gain halves as data rate doubles), and discuss the practical implication this has for a network trying to support both voice and higher-rate data services on the same infrastructure.
20. A CDMA system has 10 W of total received power at the base station and a target user's signal contributes 15 mW. Derive the SIR in both linear and dB form, showing the full calculation. Discuss what this SIR value implies about the number of other users likely sharing the channel, and why this shared-power model is fundamentally different from the orthogonal-channel isolation used in FDMA/TDMA.
21. Derive the maximum number of CDMA users supportable for a processing gain of 180 and a required linear SIR of 7, then derive the percentage change in maximum users if the required SIR rises to 10, showing both calculations. Discuss why raising the SIR requirement (e.g., to improve call quality) necessarily reduces capacity, and describe this as a defining design trade-off CDMA operators must manage continuously.
22. Derive the resulting change in SIR (in linear terms) when interference power increases by 4.5 dB, showing the dB-to-linear conversion and resulting SIR degradation clearly. Discuss the near-far effect as one practical, common cause of such an interference spike, and explain how this derivation illustrates why fast, accurate power control is critical to CDMA system stability.
23. Derive the capacity gain achieved from 3-sector sectorization, assuming interference is split equally across sectors, starting from a base case of 45 users per omnidirectional cell. Show the full calculation for the sectorized capacity. Discuss why this capacity gain in CDMA works somewhat differently from

sectorization gain in FDMA/TDMA, given CDMA's interference-limited (rather than channel-limited) nature.

24. A CDMA cell has 30 active users at a voice-activity factor of 0.4. Derive the effective number of interfering users, and then derive the resulting SIR assuming a processing gain of 200, showing both calculations. Discuss how this two-step derivation (activity factor → effective users → SIR) illustrates why CDMA is often described as having "soft capacity" rather than a fixed hard limit.
25. Derive the number of users supportable for a processing gain of 150 and an Eb/N0 requirement of 8 dB, first converting Eb/N0 to a linear SIR value before applying the capacity formula. Show every conversion step clearly. Discuss the sensitivity of your result to small changes in the Eb/N0 requirement, and why this sensitivity makes Eb/N0 optimization a high-priority focus in CDMA/3G system design.
26. Derive the change in processing gain when a user's data rate is upgraded from 9.6 kbps to 14.4 kbps at a fixed chip rate of 1.2288 Mcps, showing both before-and-after processing-gain values. Discuss the resulting effect on overall system capacity, explaining why higher-rate users consume more of the shared "capacity budget" than lower-rate voice users in a mixed-service network.
27. Derive the SIR for a system supporting 25 users with a processing gain of 256, then compare this derived SIR against a 9 dB Eb/N0 requirement (converted to linear) to determine whether the system meets its quality target. Show all conversions and the final comparison clearly. Discuss what actions an operator would take if the derived SIR fell short of the requirement.
28. A CDMA network's near-far problem causes a near user's received power to be 20 dB higher than a far user's. Derive the resulting SIR degradation experienced by the far user, showing the dB arithmetic clearly. Explain in detail how power control resolves this problem by equalizing received power levels, and discuss why this derivation demonstrates the necessity of fast, closed-loop power control in CDMA.
29. Derive the total OFDMA bandwidth for 1200 subcarriers at 15 kHz spacing, then repeat the derivation for 30 kHz spacing, showing both calculations clearly. Compare the two resulting bandwidths and explain the latency-vs-bandwidth trade-off: wider spacing shortens symbol duration (lower latency) but consumes more spectrum for the same subcarrier count. Discuss which use case (eMBB vs. URLLC) favors each option.
30. A 5G NR system uses 60 kHz subcarrier spacing with 132 resource blocks, each containing 12 subcarriers. Derive the total occupied bandwidth step by step. Discuss why 5G NR supports multiple subcarrier-spacing "numerologies" rather than the single fixed spacing used in LTE, and explain what deployment scenario this particular 60 kHz configuration is best suited for.
31. Derive the number of subcarriers needed to exactly occupy 25 MHz of bandwidth using 15 kHz subcarrier spacing, showing the full calculation. Discuss whether the result divides evenly, and if not, explain how the leftover spectrum is handled as guard band at the channel edges. Describe the purpose this guard band serves in preventing interference with adjacent frequency allocations.

32. Derive the OFDM symbol duration for subcarrier spacings of 15 kHz and 30 kHz, using symbol duration ≈ 1/spacing, showing both calculations clearly. Discuss the impact of each resulting symbol duration on multipath resilience — explaining why a longer symbol duration (narrower spacing) better tolerates delay spread, while shorter symbol duration (wider spacing) is more vulnerable to inter-symbol interference from multipath.
33. A 5G system needs to support 400 Mbps aggregate throughput using 273 resource blocks at 30 kHz subcarrier spacing. Derive the total available bandwidth from the resource-block configuration, then derive the required spectral efficiency (bps/Hz) to hit the 400 Mbps target. Discuss whether this required efficiency is realistic given typical modulation and MIMO configurations used in commercial 5G deployments.
34. Derive the total OFDMA capacity in Mbps for a system where each of 1000 subcarriers supports 4 bits/symbol at 15 kHz subcarrier spacing, using throughput ≈ subcarriers × spacing × bits/symbol, showing the full calculation. Discuss how this formula reveals three independent levers (subcarrier count, spacing, modulation order) a network can adjust to increase throughput, and the practical limits on each.
35. Derive the total number of GSM voice channels available for a 17.4 MHz allocation with 200 kHz spacing and 8 slots per carrier, after reserving 2 carriers exclusively for signaling. Show the full calculation from raw spectrum through to net voice channels. Discuss why real deployments always fall short of the theoretical maximum channel count once signaling reservations are factored in.
36. A GPRS system uses 5 time slots at 21.4 kbps each. Derive the total theoretical data rate from slot bonding, then derive the effective data rate after a 12% protocol-overhead deduction, showing both calculations clearly. Discuss the sources of this overhead (headers, retransmissions, coding schemes) and why effective throughput is consistently lower than the theoretical maximum in real deployments.
37. Derive the chip rate required to support a spreading factor of 512 at a target data rate of 7.5 kbps in a UMTS/WCDMA system, showing the calculation clearly. Discuss why UMTS instead fixes the chip rate at 3.84 Mcps and varies spreading factor to support different service rates, and explain why this design gives UMTS more flexibility than GSM's rigid fixed-slot structure.
38. A GSM carrier of 200 kHz is reused with cluster size N = 4 across a total 20 MHz spectrum. Derive the number of carriers per cell, and then the total voice channels per cell (8 slots/carrier), showing the full calculation chain. Discuss how this result would change under a larger cluster size like N = 7, and the resulting capacity-versus-interference trade-off.
39. Derive the throughput achieved by High-Speed Circuit-Switched Data (HSCSD) when 4 time slots of 14.4 kbps each are bonded for a single user, and compare this against a standard single-slot GSM data call, showing both values clearly. Discuss the capacity trade-off this bonding approach creates — a single HSCSD user consumes multiple traffic channels — and why this approach was eventually superseded by packet-switched technologies.

40. Derive the number of GSM carriers required to support 750 simultaneous voice users at 8 slots per carrier, accounting for 1 signaling slot reserved per carrier (leaving 7 usable traffic slots), showing the full calculation. Discuss why this signaling-slot reservation must be factored into capacity planning from the outset, and the consequence of underestimating required carrier count during real network design.